



CENTRE OF EXCELLENCE FOR ROAD SAFETY, IITMADRAS

ROAD SAFETY HACKATHON 2026

**Team Name:
Flamboyance**

Anuj Kumar Ranjan
Sarthak Shrivastava

IIT Kharagpur



Overview



Technical Flow



Eng

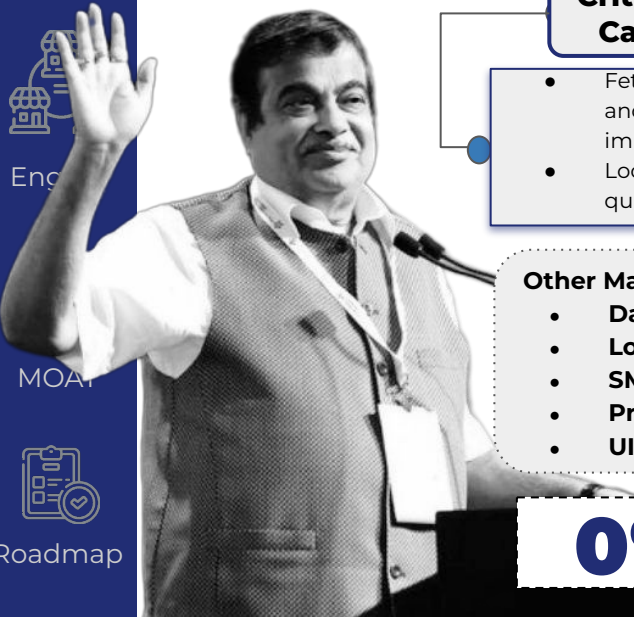
MOA



Roadmap



GLOBAL
ROAD SAFETY
PARTNERSHIP



Overview

Global Scenario	#1 Youth Killer	1.19 Million Annual Fatalities	1.19 Million Lives Lost	3260 deaths Per Day
India Figures	Remains Road Accident Capital	4.8 Lakh Accidents in 2025	1.72 Lakh Lives Lost	546 deaths per day Due to Road Accidents

Golden Hour:
60 minutes determining survival.

Key Reveals

Critical Care



- Fetches nearby hospitals and trauma centres for immediate care.
- Locates police stations for quick action.

Drive Mode



- Tracks real-time location via Drive Mode.
- Broadcasts automated SOS SMS with coordinates.

Offline First



- Works offline using seeded local databases.
- Calculates proximity without needing internet access.

Other Major Announcements:

- Database Service:** Queries local SQLite databases entirely offline without latency.
- Location Management:** Computes spatial proximity vectors locally without cloud APIs.
- SMS Service:** Broadcasts native cellular SOS alerts with exact coordinates instantly.
- Profile Management:** Persists critical emergency contacts and medical data locally safely.
- UI Components:** Implements intuitive, fluid React Native layouts for high-stress accessibility.

0%

Internet Required

100

Offline Local Queries

0ms

External API Latency



Technical Flow

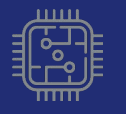
Overview



Technical Flow



Engine

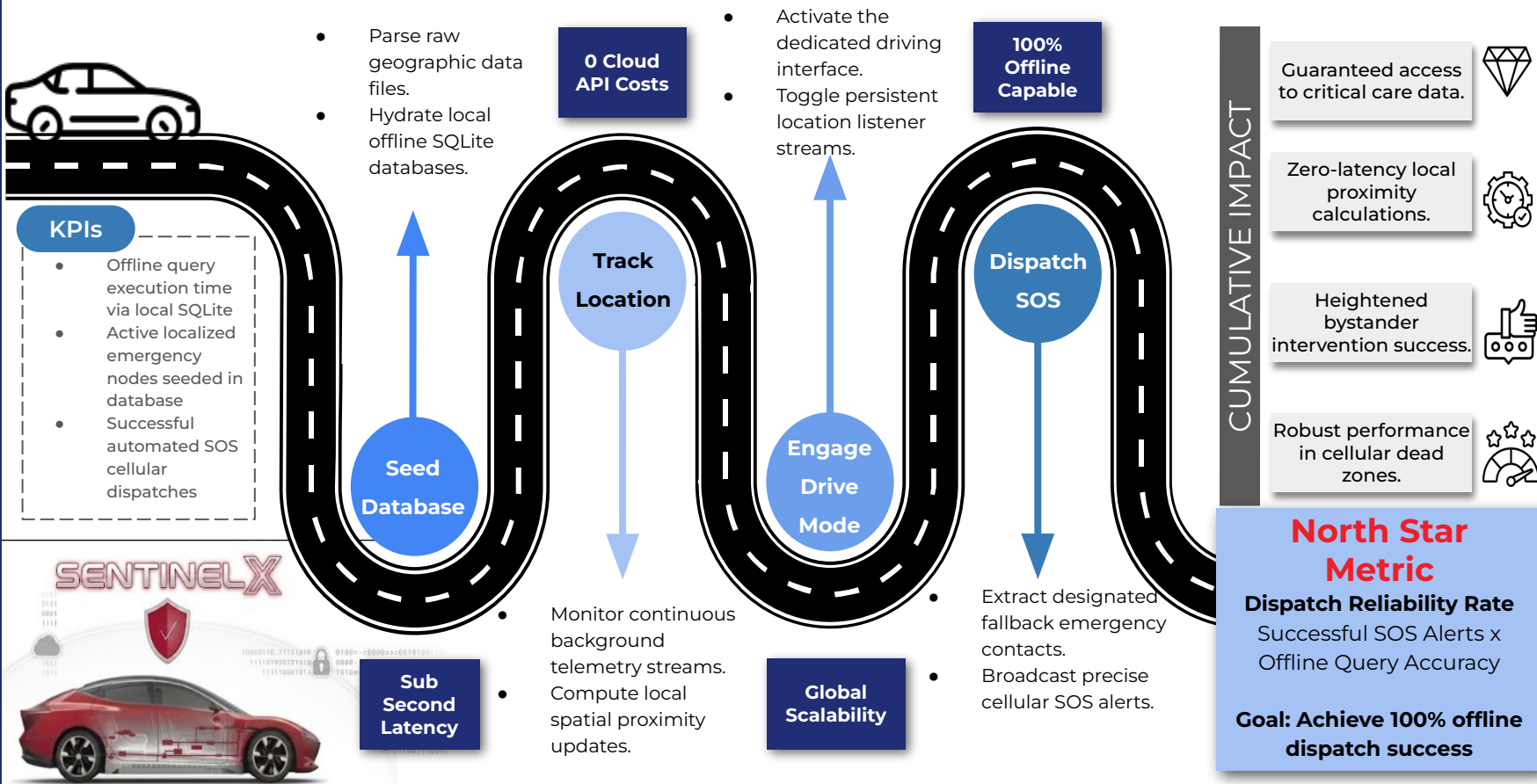


MOAT



Roadmap

SENTINEL X : Redefining Security





Overview



Technical
Flow



Engine



MOAT



Roadmap

The SentinelX Engine



Architecture



Execution



Validation

Data Resilience



Embedded
Smart Vault

- Utilizes a lightweight, localized SQLite database to bypass cloud dependency.
- Aggressively pre-caches regional emergency nodes (hospitals, police, mechanics) before entering low-bandwidth corridors.
- Guarantees instantaneous 0-Kbps data routing upon complete cellular grid failure.

Spatial Awareness



Geospatial
Telemetry

- Customized, high-contrast dark-theme mapping engine engineered for minimal visual fatigue.
- Bypasses cellular triangulation by locking directly onto satellite GPS coordinates for true offline tracking.
- Renders dynamic, spatial "Secure Zone" perimeters to visually guide users to safety.

Crisis Management



Tactical
Protocol Matrix

- Interactive, shuffling database of high-stress driving and survival SOPs (Standard Operating Procedures).
- Designed to prevent cognitive overload with a minimal-distraction, swipe-based interface.
- Instantly surfaces context-relevant action plans without requiring active network requests.

Hardware Integration



Kinematic
Sensor Engine

- Interfaces directly with native hardware accelerometers to track raw XYZ G-force in real-time.
- Employs a velocity-gated memory buffer to verify high-speed collisions and eliminate daily false positives.
- Translates physical kinetic anomalies into immediate, localized SOS triggers.

Performance
Baselines

- Query Latency:** < 200ms offline database retrieval.
- Bandwidth Dependency:** 0 Kbps required for full triage.

Fail-Safe
Fallbacks

If GPS connection drops entirely due to severe topological interference (e.g., tunnels), the system autonomously locks onto the last known coordinate vector and queues the SOS packet for instant burst transmission upon network restoration.



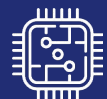
Overview



Technical
Flow



Engine



MOAT



Roadmap

System-Level Fail-Safes: The Engineering Moat

1

Kinematic Crash Detection



- Bypasses basic acceleration thresholds using a custom Velocity-Gated Memory buffer.
- Simultaneously tracks GPS speed and raw G-Force to eliminate daily false positives (e.g., dropping the phone).

2

Critical Battery Protocol



- Directly interfaces with native OS listeners to monitor precise power degradation.
- Autonomously triggers a final "last known location" SOS dispatch the millisecond power drops below 10% before device death.

3

Hardware Hijack & WakeLocks



- Overrides standard mobile OS battery optimizations during an active emergency.
- Aggressively requests OS WakeLocks to prevent screen timeout and forces maximum display brightness for high-stress visibility.

4

Spatial Rolling Cache



- Preemptively downloads localized emergency node data for the surrounding 50km radius.
- Dynamically overwrites old cache as the user travels, ensuring the offline SQLite database remains highly relevant without bloating phone storage.

5

Autonomous Execution Matrix

AUTONOMOUS TRIGGERS



> 4.0G Impact + > 30km/h Velocity = Auto-SOS Dispatch



< 10% Battery State = Final Geo-Ping Executed



SOS State Active = WakeLock + 100% Brightness Forced



0 Kbps Network Detected = Local SQLite Vault Served



Overview



Technical
Flow



Engine

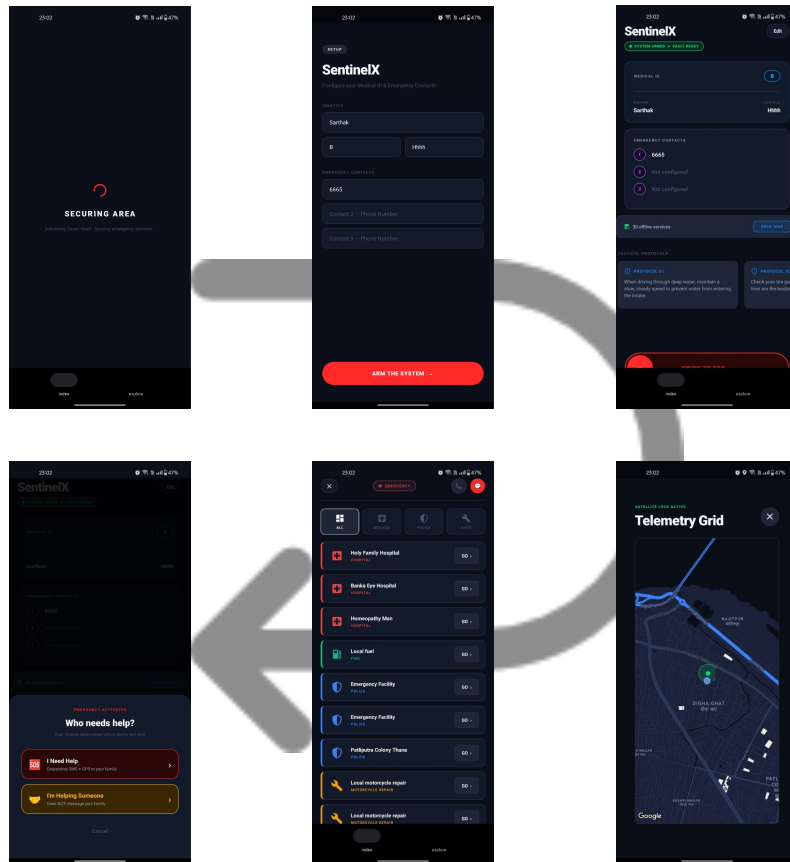


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Roadmap

Interface & Roadmap



Git Repository Link: <https://github.com/Sarthak790/RoadSoS/tree/Anuj>

Future Roadmap

P2P SOS Bouncing

- Mesh Networking: Utilizes BLE and Wi-Fi Direct protocols.
- Offline Relays: Bounces encrypted SOS packets through nearby devices in cellular dead zones.
- Decentralized Grid: Creates a peer-to-peer safety net of user relay nodes.

Wearable Data Hooks

- Smartwatch Integration: Interfaces with Apple Watch, Garmin, and WearOS.
- Biometric Sync: Feeds live vitals (heart rate, SpO2) into the Kinematic Engine.
- High-Fidelity Triggers: Differentiates minor phone drops from genuine medical emergencies.

OEM Satellite Integration

- Hardware APIs: Leverages emerging Android 15/16 direct-to-satellite capabilities.
- LEO Routing: Transmits 0-Kbps payloads directly to Low Earth Orbit satellites.
- True Global Coverage: Completely bypasses cellular and Wi-Fi dependencies.

THANK YOU !

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